



Frailty

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INTRODUCTION

Frailty is a clinical syndrome of aging-related physiological vulnerability characterized by reduced resilience to stressors and increased risk of adverse health outcomes. It reflects cumulative declines across multiple systems, resulting in impaired homeostatic reserve and recovery capacity. Clinically, frailty often presents as decreased energy, muscle weakness, slowed performance, and unintentional weight loss, even in the absence of overt disease. Recognition of frailty is essential to guide care planning, calibrate interventions, and avoid iatrogenic harm.

This topic will review the epidemiology, conceptual frameworks, biologic mechanisms, diagnostic approaches, and clinical management strategies relevant to frailty in older adults.

Failure to thrive is discussed separately. (See "[Failure to thrive in older adults: Evaluation](#)" and "[Failure to thrive in older adults: Management](#)".)

DEFINITION

Frailty is widely recognized as a clinical syndrome reflecting age-related vulnerability and diminished physiological reserve, resulting in an increased risk of adverse health outcomes when exposed to stressors such as illness, injury, or surgery [1,2]. Frailty is distinct from but often overlaps with disability and comorbidity; it is not an inevitable consequence of aging, nor does old age alone define frailty [3-7].

Frailty exists along a continuum, from robust to pre-frail to frail, with the potential for reversibility if identified early. In its most advanced form, often described as "failure to thrive," even minor stressors can lead to significant decline. (See "[Failure to thrive in older adults: Evaluation](#)" and "[Failure to thrive in older adults: Management](#)".)

Major conceptual frameworks — There are two dominant frameworks that have shaped the understanding and measurement of frailty in clinical and research settings:

- **Physical (Phenotypic) Frailty** – This model was developed primarily in community-dwelling older adults and focuses on observable signs and symptoms such as unintentional weight loss, exhaustion, weakness, slow walking speed, and low physical activity [3].
- **Deficit accumulation (Frailty Index [FI])** – This model conceptualizes frailty as the cumulative effect of a broad range of health deficits, including diseases, symptoms, and disabilities across multiple domains (physical, psychological, cognitive, and social) [8-11].

These two frameworks both identify individuals at increased risk for poor outcomes, but they often classify different individuals as frail, especially in those with intermediate frailty [8,12,13].

Key components — Key components of frailty include unintentional weight loss, fatigue or reduced energy, decreased muscle mass and strength, slowed gait or physical performance, and some degree of vulnerability in function.

EPIDEMIOLOGY

Prevalence — The prevalence of frailty varies widely depending on the population studied, the specific frailty assessment tool used, and regional or socioeconomic factors [1,14-16]. Among community-dwelling older adults, prevalence estimates using physical frailty measures (eg, Fried phenotype) typically range from 11 to 15 percent, while deficit accumulation models (eg, Frailty Index) yield estimates of 20 to 29 percent. Pre-frailty is more common, affecting 45 to 49 percent of community-dwelling older adults [17,18]. In middle-aged adults (45 to 64 years), frailty is less common, with prevalence estimates ranging from 5 to 11 percent [19-21].

A 2021 global report covering 62 countries showed that frailty is more prevalent in low-resource countries compared with high-resource countries, and it is more common among women, those with lower education, and those living alone or in poverty [16]. Studies in Asian and

Latin American countries have found prevalence rates ranging from 4 to over 20 percent, reflecting substantial heterogeneity due to differences in diagnostic criteria, age distribution, and socioeconomic status [22-24].

Chronic comorbidities, including hypertension, heart failure, diabetes, and multimorbidity (≥ 2 chronic diseases), are associated with frailty, and frailty is particularly common in patients with cancer [1,16,25-29].

Health consequences and mortality — Frailty negatively influences health outcomes among older adults. One of the pivotal aspects is its capacity to decrease resilience, manifesting as heightened vulnerability to stressors such as acute illness, surgery, or trauma [30-34]. Frailty is both a predictor and amplifier of adverse health events, making it a focus for risk stratification and intervention.

Frail older adults face significantly higher risks of:

- **Postoperative complications** – Frail patients undergoing surgery have a higher risk of 30-day postoperative mortality, infections, delirium, and extended hospital stays compared with non-frail patients [35-37]. For example, in a meta-analysis, frail patients undergoing emergency surgery had nearly a fourfold increased odds of 30-day mortality compared with non-frail patients (odds ratio [OR] 3.85, 95% CI 2.83-5.24) [38].
- **Falls and fractures** – In meta-analyses, frailty has been associated with an increased risk of falls and fractures [39-42]. Further, older frail patients who experience falls are at higher risk of prolonged hospital stays and complications [43].
- **Hospitalization and institutionalization** – Frailty is associated with an increased likelihood of unplanned hospital admissions and long-term care placements among older adults [44].
- **Mental health outcomes** – Frailty appears to influence mental health outcomes [45-47], including an increased risk of suicidal ideation and suicide attempts [48-51].
- **Progression and impact of chronic disease** – Individuals with both frailty and chronic diseases exhibit a bidirectional relationship, where frailty increases the risk of developing chronic diseases, while chronic diseases can exacerbate frailty [52]. For example, older adults with both frailty and diabetes face the highest mortality risk, compared with those with either condition alone [53]. Frailty is implicated in the accelerated progression of neurodegenerative diseases, including Alzheimer disease and vascular dementia [54,55].
- **Mortality** – Frailty increases the risk of all-cause mortality [56]. Frailty is associated with higher in-hospital mortality in older adults with acute myocardial infarction [57-62]. Across diverse populations and frailty assessment tools, adjusted hazard ratios for all-cause mortality in frail versus non-frail older adults with cancer typically range from approximately 2.1 to 2.8, even after controlling for age, cancer stage, sex, comorbidities, and cancer type [63-71]. During the coronavirus disease 2019 (COVID-19) pandemic, frailty predicted mortality more strongly than age or comorbidity alone [72-77]. In the COVID-OLD study, patients with higher Clinical Frailty Scale scores faced up to threefold greater risk of death during hospitalization compared with less frail peers [78].

PATHOPHYSIOLOGY

Frailty arises from the dysregulation of multiple interrelated physiologic systems, involving chronic inflammation, sarcopenia, hormonal imbalance, mitochondrial dysfunction, and impaired metabolic signaling. These biologic alterations, often accelerated by aging, disease, and environmental stressors, undermine physiologic reserve and increase vulnerability to minor stress ([figure 1](#)) [4]. Emerging contributors include extracellular vesicle signaling [79-83], alterations to the gut microbiome [84-91], biologic clocks [92-95], and circulating cell-free deoxyribonucleic acid (cfDNA) [96-101].

Key mechanistic pathways include:

- **Chronic inflammation** – Frailty is associated with chronic, low-grade inflammation, termed inflammaging [92,102-106], marked by elevated levels of interleukin 6 (IL-6), tumor necrosis factor alpha (TNF-alpha), C-reactive protein (CRP), and soluble TNF receptors. These cytokines result in impaired adaptive immunity, reduced vaccine efficacy, and greater susceptibility to infection [107-111], as well as hematologic effects, including anemia of inflammation [112-117], endothelial dysfunction, and thrombosis, due to increased levels of fibrinogen, D-dimer, and Factor VIII [111,118,119].
- **Sarcopenia and musculoskeletal decline** – Age-related decline in muscle mass and strength is a hallmark of frailty, driven by inflammation, hormonal changes, and mitochondrial dysfunction, as well as reduced satellite cell function and regenerative capacity with age [120-122]. Sarcopenia and frailty frequently coexist and reinforce each other [123-126].
- **Endocrine dysregulation** – Multiple age-related hormone changes have been associated with frailty. Age-related shifts in hormonal balance contribute to frailty by promoting catabolism, reducing musculoskeletal integrity, and impairing stress responses [122]. Key alterations include decreased levels of growth hormone, insulin-like growth factor 1 (IGF-1), dehydroepiandrosterone sulfate, [testosterone](#), estrogen, and vitamin D; increased cortisol; and alterations in the renin angiotensin system. Clinical correlates include decreased strength, balance, mobility, muscle mass, and immune function, as well as greater bone loss and fall risk [127-150].

- **Mitochondrial dysfunction** – Mitochondrial dysfunction is increasingly recognized as a central contributor to frailty in older adults, resulting in faster depletion of muscle energy stores during exertion, contributing to decreased physical performance and increased fatigability [151-156]. Measurement of mitochondrial respiration may serve as a minimally invasive biomarker for frailty [154,157].
- **Altered nutrient sensing and metabolism** – Disruption in nutrient sensing and metabolic regulation is increasingly recognized as a core mechanism contributing to frailty [158,159]. Pathways such as mechanistic target of rapamycin (mTOR), adenosine monophosphate-activated protein kinase (AMPK) [1,160], and the insulin/IGF-1 [161] axis are crucial regulators of cellular energy balance, anabolic signaling, and stress responses. These pathways modulate the body's ability to adapt to nutritional and energetic stressors, and their impairment can promote catabolism, leading to frailty [162].

MEASUREMENT TOOLS

A wide array of frailty measurement tools have been validated, capturing different domains and offering unique advantages and limitations [1,163-167]. Most tools reliably identify individuals at highest risk for adverse outcomes, but their agreement and predictive performance can vary depending on the population and context [165,168-171].

Choosing a frailty assessment tool — When choosing a frailty assessment tool, clinicians should consider the care setting, time constraints, and intended use [163].

- **Routine clinical settings** – There are several brief, validated screening tools suitable for primary care and community settings, with high feasibility and diagnostic accuracy (see '[Rapid screening tools](#)' below). More detailed instruments, such as the Fried Frailty Phenotype and the Frailty Index (FI), are valuable in comprehensive evaluations. (See '[Fried Frailty Phenotype](#)' below and '[Frailty Index](#)' below.)
- **Prior to treatments or interventions** – When frailty assessment is used to inform high-stakes treatment decisions (eg, chemotherapy or surgery), tools validated in those specific populations may offer more accurate predictions than general-purpose tools [1]. [eFrailty.org](#) provides a helpful guide for selecting appropriate frailty instruments, along with electronic calculators for commonly used tools.

Fried Frailty Phenotype — One of the most widely used physical frailty tools is the Fried Frailty Phenotype, which defines frailty based on five criteria [3]. Individuals meeting three or more criteria are classified as frail, one or two as pre-frail, and none as robust ([table 1](#)):

- Weight loss (≥ 5 percent of body weight in last year)
- Exhaustion (positive response to questions regarding effort required for activity)
- Weakness (decreased grip strength)
- Slow walking speed (gait speed) (>6 to 7 seconds to walk 15 feet)
- Decreased physical activity (Kcals spent per week; males expending <383 Kcals and females <270 Kcals)

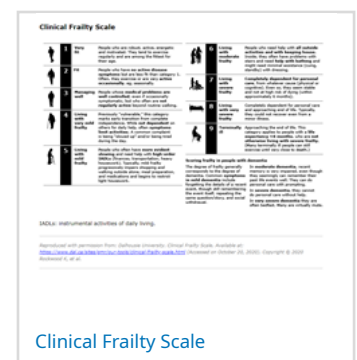
This tool, validated in the Cardiovascular Health Study (CHS), which involved over 5000 adults aged ≥ 65 years [3], is predictive of adverse clinical outcomes but requires direct physical measurements, which may limit its feasibility in some settings.

Frailty Index — The FI quantifies frailty as a ratio of the number of deficits present to the total number assessed [172]. It requires answering 20 or more medical and function-related questions. The higher the number of deficits, the higher the frailty score. This multidimensional tool can be adapted to information available in medical records and does not require patient examination. The FI's predictive validity for mortality and hospitalization is well established [10,173-180].

Rapid screening tools — Although the Fried Frailty Phenotype assessment and the FI approach are the most commonly cited and most validated approaches to frailty measurement, clinicians and patients may benefit from using a quicker frailty screening assessment tool. Studies comparing the most commonly utilized screening tools indicate that rapid screening tools are comparable to other tools in predicting risk of adverse health outcomes and mortality [181-183].

Some examples include:

- **Clinical Frailty Scale (CFS)** – A global clinical judgment-based tool that grades frailty from 1 (very fit) to 9 (terminally ill) ([figure 2](#)) [184]. In acute care and surgical environments, the CFS is widely used for rapid risk stratification. Higher CFS scores are independently associated with increased risk of mortality, longer hospital and intensive care unit (ICU) stays, institutional discharge, complications (such as delirium), and new disability after surgery, even after adjusting for age and illness severity [167,185,186].
- **Edmonton Frail Scale (EFS)** – The EFS has 14 questions that cover a variety of domains, including cognition, general health, function, social support, and nutrition [187-189]. This tool, while not diagnosing physical frailty or deficit accumulation frailty per se, is useful to identify a number of measurement domains that are important to the health care of older adults [190].
- **PRISMA-7 screening tool** – This is a brief, seven-item questionnaire designed to identify frailty in older adults. It consists of yes/no questions addressing age, sex, health problems, need for



regular help, mobility, and social support. A score of three or more is commonly used as the threshold to indicate possible frailty, though some studies suggest a cutoff of four may improve specificity in certain populations [191-193].

- **FRAIL Scale** – This is a brief questionnaire suitable for community and primary care settings, with good feasibility and reliability [166,168,194-196]. However, as with other tools that include mobility items, clinicians should interpret scores cautiously in people whose difficulty walking or climbing stairs is primarily due to focal musculoskeletal or neurologic problems (eg, advanced osteoarthritis, amputation), as this may lead to misclassification.

The mnemonic "FRAIL" is helpful in remembering the component questions:

- **Fatigue** ("Have you felt fatigued? Most or all of the time over the past month?") – Yes = 1, No = 0
- **Resistance** ("Do you have difficulty climbing a flight of stairs?") – Yes = 1, No = 0
- **Ambulation** ("Do you have difficulty walking one block?") – Yes = 1, No = 0
- **Illnesses** ("Do you have any of these illnesses: hypertension, diabetes, cancer (other than a minor skin cancer), chronic lung disease, heart attack, heart failure, angina, asthma, arthritis, stroke, and kidney disease?") – Five or greater = 1, fewer than 5 = 0
- **Loss of weight** ("Have you lost more than 5 percent of your weight in the past year?") – Yes = 1, No = 0

FRAIL Scale scores range from 0 to 5 (0 = best, 5 = worst) and represent frail (3 to 5), pre-frail (1 to 2), and robust (0) health status.

DIAGNOSTIC APPROACH

Early identification of frailty in clinical practice is essential for guiding further assessment, tailoring interventions, and optimizing outcomes for older adults. The diagnosis is made based on a combination of history, physical examination findings, and the use of a frailty scale. When applying most frailty instruments, clinicians are advised to rate a person's usual health state over the preceding weeks to months, rather than a transient decline during an acute illness or hospitalization.

History — Key elements of the history include the following, many of which are included in screening instruments:

- **Energy and fatigue** – Assess for new or worsening fatigue, reduced stamina, and low activity levels.
- **Functional status** – Inquire about ability to perform activities of daily living (ADLs) and instrumental ADLs (IADLs), as well as stair climbing and walking outside the home. (See "[Office-based assessment of the older adult](#)", [section on 'Components of functional status'](#).)
- **Weight loss** – Document unintentional weight loss (quantified as ≥ 5 percent body weight over 12 months).
- **Frailty screen** – Completion of a frailty screening tool such as the FRAIL Scale, EFS, CFS, or PRISMA-7 should be part of the evaluation. (See '[Rapid screening tools](#)' above.)

Physical examination — The physical examination should be comprehensive and include evaluation of any specific symptoms, including those of unintentional weight loss or fatigue. (See "[Approach to the patient with unintentional weight loss](#)", [section on 'Physical examination'](#)" and "[Approach to the adult patient with fatigue](#)", [section on 'Physical examination'](#).)

If a frailty screening tool performed during the initial history suggests frailty, a full frailty assessment should be conducted using a validated instrument.

The physical examination should also include the following, some of which are components of the physical frailty assessment tool ([table 1](#)) (see '[Fried Frailty Phenotype](#)' above):

- Gait speed test (timing a 4-meter walk; <0.8 meters/second is highly sensitive for frailty)
- Chair rise test (timing five consecutive stands from a chair without using arms; taking longer than 12 to 15 seconds, or being unable to complete the task, suggests frailty)
- Handgrip strength using a dynamometer, if feasible

Laboratory testing — Essential laboratory tests include a complete blood count (CBC), basic metabolic panel (BMP), liver function tests, thyroid-stimulating hormone (TSH), vitamin B12, folate, 25-hydroxyvitamin D, and albumin. These tests help identify reversible contributors such as anemia, kidney or hepatic dysfunction, hypothyroidism, vitamin deficiencies, and malnutrition, all of which are common and potentially modifiable in frail older adults. Additional tests may include fasting glucose or HbA1c to screen for diabetes, and urinalysis if indicated by symptoms [197,198].

Further testing should be guided by clinical suspicion, including appropriate workup for suspected infections or malignant processes. Cardiac studies (eg, N-terminal prohormone of brain natriuretic peptide [NT-proBNP], high-sensitivity troponin, electrocardiogram [ECG],

echocardiogram) should be considered if there are symptoms or signs of heart failure, arrhythmia, or unexplained dyspnea. Rheumatologic tests are warranted if there is clinical suspicion for polymyalgia rheumatica, vasculitis, or other connective tissue diseases.

Conditions that contribute to frailty

- **Cognition and mood** – Changes in memory, attention, or executive function and depressive symptoms may exacerbate frailty. (See ["Mental status scales to evaluate cognition"](#) and ["Screening for depression in adults"](#), section on 'Screening tests'.)
- **Sleep disturbances** – Sleep disturbances are common in older adults and may contribute to fatigue, cognitive decline, and worsening frailty [199,200]. As part of the frailty assessment, clinicians should routinely ask about sleep quality, duration, nighttime awakenings, excessive daytime sleepiness, and use of sleep aids. Validated tools such as the Pittsburgh Sleep Quality Index (PSQI) are recommended to screen for sleep quality, duration, insomnia symptoms, and daytime dysfunction, as poor sleep quality and abnormal sleep duration (short <5 hours or long >8 to 9 hours) are independently associated with frailty and pre-frailty.
- **Social support and recent changes** – Social isolation, recent hospitalizations, or changes in living situation can affect frailty and may affect the treatment plan.

DIFFERENTIAL DIAGNOSIS

Frailty is a clinical syndrome, not a diagnosis of exclusion. However, it is critical to consider reversible or treatable conditions that can mimic or contribute to frailty. These include:

- **Occult malignancy** – Unintentional weight loss, anemia, and constitutional symptoms should prompt evaluation for underlying cancer (eg, lymphoma, myeloma, solid tumor). Patients should have all age-appropriate cancer screening. (See ["Overview of preventive care in adults"](#), section on 'Cancer screening'.)
- **Depression** – Depression can present with fatigue, weight loss, cognitive complaints, social withdrawal, and functional decline, all of which can mimic or exacerbate frailty. Symptoms such as persistent low mood, feelings of worthlessness, hopelessness, guilt, or anhedonia may help distinguish it from frailty alone. (See ["Approach to the adult patient with suspected depression"](#).)
- **Heart failure** – Heart failure can also present with fatigue, reduced exercise tolerance, weight changes, and functional decline. Symptoms such as dyspnea, orthopnea, and peripheral edema may help distinguish it from frailty alone. (See ["Approach to diagnosis and evaluation of acute decompensated heart failure in adults"](#).)
- **Chronic kidney disease** – The condition may cause fatigue, poor appetite, weight loss, cognitive changes, and functional decline, especially in advanced stages. Laboratory evaluation of kidney function is essential to the diagnosis. (See ["Chronic kidney disease \(newly identified\): Clinical presentation and diagnostic approach in adults"](#).)
- **Hypothyroidism** – Hypothyroidism can present with fatigue, weight gain or loss, cognitive slowing, and functional impairment, often with subtle or nonspecific symptoms in older adults. Laboratory evaluation is essential to the diagnosis. (See ["Diagnosis of and screening for hypothyroidism in nonpregnant adults"](#).)
- **Anemia** – Anemia can cause fatigue, weakness, and exercise intolerance. Etiologies include iron, B12, or folate deficiency, chronic disease, or myelodysplasia. (See ["Diagnostic approach to anemia in adults"](#).)
- **Malnutrition/vitamin deficiencies** – Protein-energy malnutrition and deficiencies in vitamins (A, D, E, B12, C) are common in older adults and can cause or worsen frailty, cognitive decline, and functional impairment [201]. (See ["Nutritional issues in older adults"](#).)
- **Parkinson disease** – Parkinson disease may present with bradykinesia, rigidity, tremor, gait disturbance, and nonmotor symptoms such as sleep dysfunction, depression, and cognitive decline. (See ["Diagnosis and differential diagnosis of Parkinson disease"](#).)
- **Other neurodegenerative disorders** – Disorders such as dementia with Lewy bodies, frontotemporal dementia, and amyotrophic lateral sclerosis can contribute to frailty through motor, cognitive, and behavioral symptoms. (See ["Clinical features and diagnosis of dementia with Lewy bodies"](#) and ["Frontotemporal dementia: Clinical features and diagnosis"](#) and ["Clinical features of amyotrophic lateral sclerosis and other forms of motor neuron disease"](#).)
- **Systemic vasculitis/polymyalgia rheumatica** – These conditions can present with fatigue, weight loss, proximal muscle pain/stiffness, and elevated inflammatory markers. Testing of erythrocyte sedimentation rate (ESR), C-reactive protein (CRP), and autoantibodies may be helpful. (See ["Overview of and approach to the vasculitides in adults"](#), section on 'Diagnostic approach' and ["Clinical manifestations and diagnosis of polymyalgia rheumatica"](#), section on 'Diagnostic approach'.)
- **Acute cerebrovascular event (stroke/transient ischemic attack)** – Sudden onset of focal neurologic deficits, cognitive changes, or functional decline should prompt urgent neuroimaging and evaluation. (See ["Initial evaluation and management of transient ischemic attack and minor ischemic stroke"](#).)

MANAGEMENT

Multicomponent approach — Management of frailty should be individualized and multidisciplinary, addressing physical, nutritional, psychologic, and social domains. Effective care includes both structured interventions and practical, common-sense approaches that can be implemented across settings, with an awareness of goals of care ([figure 3](#)).

Multicomponent interventions that combine exercise, nutrition, cognitive training, and social support appear to yield the best outcomes for frailty prevention and management [202-204]. In particular, meta-analyses have shown that combining exercise with nutritional support yields benefits in reducing frailty among older adults [205,206]. Multicomponent interventions that also incorporate cognitive training, psychosocial support, and medication optimization have demonstrated broader efficacy in mitigating frailty-related outcomes [207].

General measures in management — The following measures are appropriate for all frail and pre-frail adults [202,208-211].

- **Medication review** – Assess medications for appropriateness (using tools such as the Beers, STOPP, and START criteria) and deprescribe agents contributing to sedation, falls, orthostasis, or cognitive impairment [212]. Deprescribing for patients with frailty is safe and can improve outcomes [213]. (See "[Drug prescribing for older adults](#)", [section on 'Potentially inappropriate medications'](#) and "[Deprescribing](#)", [section on 'Deprescribing specific medications'](#).)
- **Encourage social engagement and support** – Promote and facilitate community engagement, caregiver support, and referral to local services.
- **Health education** – Teach older adults and caregivers about frailty, mobility strategies, fall prevention, and nutrition.
- **Prioritize functional preservation** – Treat chronic conditions, promote physical activity, and provide preventive care. Encourage exercise, particularly resistance and aerobic training, yoga, or tai chi, to improve mobility, muscle strength, and balance. Engage occupational and physical therapy to address specific deficits and promote safe mobility and activities of daily living (ADLs).
- **Improve sleep to support function** – Improving sleep can enhance energy, mood, and physical performance in frail older adults. However, pharmacologic options are limited due to fall risk and other adverse effects (see "[Overview of the treatment of insomnia in adults](#)", [section on 'Older adults'](#)). Instead, prioritize nonpharmacologic approaches ([table 2](#)). (See "[Insufficient sleep: Evaluation and management](#)", [section on 'Counseling on lifestyle changes'](#).)
- **Monitoring** – Mobile health interventions, such as wearable sensor technologies, can track frailty-related metrics such as gait speed, strength, and weight changes in real time, which may enable timely clinical interventions [214].

Comprehensive geriatric assessment — A comprehensive geriatric assessment (CGA) is essential to providing a complete assessment of function and prognosis and can guide the development of management and intervention plans. (See "[Comprehensive geriatric assessment](#)".)

Goals of care discussion — A goals of care discussion is important for all older adults, particularly those with frailty (see "[Discussing goals of care](#)"). For example, screening or intervention for some conditions may not provide an appreciable benefit, and some procedures or hospitalizations may pose an unnecessary burden [215]. In some cases, referral to palliative care may be the most appropriate intervention. (See '[Palliative care](#)' below.)

However, many frail older adults may still benefit from some interventions, including those targeting specific components of their frailty examination. Input from a multidisciplinary team evaluating medical, functional, cognitive, psychologic, and social issues, followed by a coordinated care plan, is often the best way to prioritize care for frail older adults with multimorbidity [202,216].

Palliative care — Diagnosing frailty helps identify patients at highest risk of adverse health outcomes. Palliative care approaches may be helpful to alleviate symptoms and to help determine the appropriateness of potential medical and surgical interventions. For those with advanced frailty and multiple comorbidities or those with failure to thrive, the engagement of a palliative care team may help to identify methods to preserve or improve quality of life and to clarify goals of care. (See "[Benefits, services, and models of subspecialty palliative care](#)", [section on 'Palliative care services'](#).)

Exercise — Structured, multicomponent exercise programs (eg, including strength, balance, aerobic, flexibility) are appropriate interventions for both frailty prevention and slowing of frailty progression, similar to older adult populations in general (see "[Physical activity and exercise in older adults](#)", [section on 'Overview of physical activity components'](#)). Group-based and supervised programs show benefit in improved mobility, muscle strength, and reduced falls [203,204,217]. However, long-term outcomes are uncertain.

In a 2024 meta-analysis of 28 trials involving 4857 older adults, structured multicomponent exercise programs of at least 12 weeks' duration led to measurable improvements in both frailty status and physical performance [218]. However, the benefits were modest, and in most trials, either participants or study personnel were not blinded.

In a systematic review of 33 trials conducted among adults over 65 years of age, yoga improved the frailty markers of gait speed, lower-extremity strength, and endurance when compared with education or inactive controls [219]. Yoga outcomes were similar to those seen with

the active controls of tai chi or other forms of exercise. (See ["An overview of yoga and its health benefits and risks"](#).)

Although patients with frailty may not be able to meet minimum recommended activity levels for older adults, even modest activity and muscle strengthening can impact the progression of functional limitations [220,221]. (See ["Physical activity and exercise in older adults"](#), section on 'Functionally limited or frail'.)

In addition to exercise, formal occupational therapy has been shown to be helpful, particularly for patients who are having difficulties with ADLs. A meta-analysis of home and community-based occupational therapy for frail older adults showed moderate improvements in the ability to perform ADLs, in social participation, and in mobility, compared with baseline [222].

Nutrition — Systematic reviews have shown inconsistent effects of nutritional strategies in improving frailty status or physical performance [223-225]. For patients with weight loss, attention should be focused on causes of malnutrition, which may include medication side effects, depression, difficulties with chewing and swallowing, dependency on others for eating, and adherence to unnecessary dietary restrictions (low-salt/low-fat) ([table 3](#)).

Although oral nutritional supplements between meals (low-volume, high-calorie drinks or puddings) may help add protein and calories, their overall benefit is limited. (See ["Failure to thrive in older adults: Management"](#), section on 'Weight loss'.)

Vitamin D supplementation in those with low levels may reduce falls, improve balance, and preserve muscle strength. This is discussed separately. (See ["Falls: Prevention in community-dwelling older persons"](#) and ["Geriatric health maintenance"](#), section on 'Vitamin D'.)

Limited role for pharmacologic interventions — No pharmacologic agent is established as a first-line treatment for frailty, and most interventions have shown only modest benefits in clinically meaningful outcomes [226-228]. Ineffective therapies include estrogen therapy, growth hormone therapy, and dehydroepiandrosterone sulfate (DHEA-S) supplementation [229-231]. [Testosterone](#) replacement may improve muscle mass or strength in males with hypogonadism, but routine use is not recommended due to limited efficacy and potential adverse effects. This is discussed separately. (See ["Testosterone treatment of male hypogonadism"](#), section on 'Muscle strength/fat-free mass'.)

Some studies suggest that angiotensin-converting enzyme inhibitors and angiotensin receptor blockers may improve muscle strength and physical function, but results are inconsistent and insufficient to support routine use [232-235]. Emerging studies suggest that [metformin](#) may reduce inflammation, cellular senescence, and muscle fibrosis, with early trials in older adults showing improved muscle recovery after disuse. Lactoferrin and other anti-inflammatory or immunomodulatory compounds are under investigation, but robust clinical data are lacking [236-238].

MODELS OF CARE

An interdisciplinary team-based approach to care is often best at meeting the needs of frail adults. However, outcomes of interdisciplinary interventions are mixed. The frailest older adults in the United States may benefit from the Medicare Program of All-inclusive Care for the Elderly (PACE). PACE was designated as a permanent Medicare program in 1997 because it was successful in maintaining function and community-dwelling status and in providing lower-cost care compared with traditional fee-for-service care [239]. The goals of this model are to improve function, overcome environmental challenges, and keep older adults living in their communities by preventing institutionalization.

In this outpatient patient management model, primary care is delivered to community-dwelling older adults by an interdisciplinary team in a day clinic setting [240]. The interdisciplinary team includes a practitioner knowledgeable in geriatric medicine, nurses, physical and occupational therapists, and social workers. Services provided include home nursing, physical and occupational therapy, transportation, aide service, and adult day care. Patients who enter the program receive complete long-term care and are followed until the end of their lives, even if they eventually require placement in an assisted-living facility or nursing home.

SUMMARY AND RECOMMENDATIONS

- **Definition** – Frailty is a multidimensional clinical syndrome of diminished physiologic reserve and impaired stress response, resulting in increased risk of adverse health outcomes, including falls, hospitalization, cognitive decline, and mortality. While more common with age, frailty is distinct from aging alone, and its presence varies widely among older adults. (See ["Definition"](#) above.)
- **Epidemiology and health consequences** – The prevalence of frailty increased with age and is more common in patients with comorbid conditions. Health consequences are diverse and include a greater risk of postoperative complications, fractures, institutionalization, and mortality. (See ["Epidemiology"](#) above.)
- **Pathophysiology** – Frailty arises from the interplay of chronic inflammation, hormonal dysregulation, mitochondrial dysfunction, sarcopenia, and altered nutrient sensing. These mechanisms collectively impair resilience and contribute to progressive decline. (See ["Pathophysiology"](#) above.)

- **Measurement** – A wide array of frailty measurement tools has been validated, capturing different domains and offering unique advantages and limitations. Rapid screening tools are useful, although more detailed tools may be appropriate in some settings. (See ['Measurement tools'](#) above and ['Choosing a frailty assessment tool'](#) above.)
- **Diagnosis** – Early identification of frailty in clinical practice is essential for guiding further assessment, tailoring interventions, and optimizing outcomes for older adults. Diagnosis rests on focused history (fatigue, function, nutrition, mood, cognition), targeted physical examination (eg, chair stand, gait speed), and limited laboratory testing. The use of validated screening tools, such as the FRAIL Scale or Clinical Frailty Scale (CFS), may be helpful. (See ['Diagnostic approach'](#) above.)
- **Management**
 - Frailty management should begin with a comprehensive geriatric assessment (CGA) and goals of care discussion. Palliative care may be most appropriate for patients with advanced frailty. (See ['Comprehensive geriatric assessment'](#) above and ['Goals of care discussion'](#) above and ['Palliative care'](#) above.)
 - Measures suitable for all patients include medication optimization, supportive care, and nutrition optimization. (See ['General measures in management'](#) above and ['Nutrition'](#) above.)
 - A structured multicomponent exercise program and optimal nutrition are important components of treatment. (See ['Management'](#) above.)

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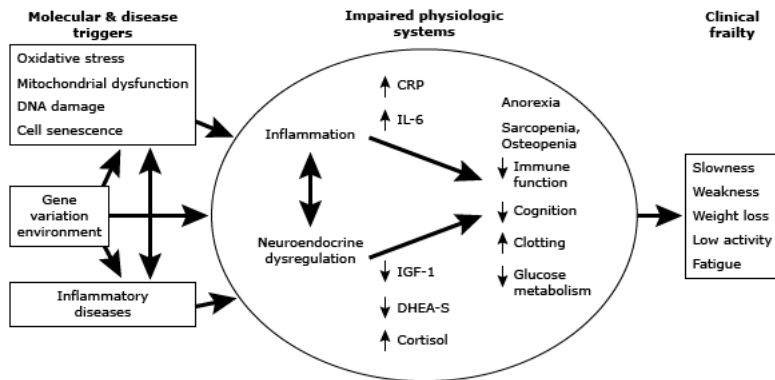
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Topic 3010 Version 62.0

GRAPHICS

Figure 1: Hypothesized model of frailty and adverse health outcomes



CRP: C-reactive protein; IL: interleukin; IGF: insulin-like growth factor; DHEA-S: dehydroepiandrosterone sulfate.

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Graphic 60373 Version 3.0

Table 1: Physical frailty phenotype

Characteristic	Measurement	Meets criteria if:			
Shrinking - Unintentional weight loss - Sarcopenia (loss of muscle mass)	Weight changes	>10 lbs or 5% body weight unintentional weight loss in the past year			
		or BMI <18.5 kg/m ²			
Weakness	Grip strength <i>(requires hand dynamometer)</i>	Lowest 20% (by sex, BMI)			
		Males:		Females:	
		BMI	Cutoff for grip strength (Kg) criterion for frailty	BMI	Cutoff for grip strength (Kg) criterion for frailty
		≤24	≤29	≤23	≤17
		24.1 to 26	≤30	23.1 to 26	≤17.3
		26.1 to 28	≤30	26.1 to 29	≤18
		>28	≤32	>29	≤21
Poor endurance Exhaustion	Self-report of exhaustion	Subjective patient report to following: I felt that everything I did was an effort (Answers: yes, no, refused, don't know) If yes, how often in the last week did you feel this way? 3 - Most of the time 2 - Moderate amount of time (3 to 4 days) 1 - Some of the time (1 to 2 days) 0 - Rarely/none of the time (<1 day) I could not get going (Answers: yes, no, refused, don't know) If yes, how often in the last week did you feel this way? 3 - Most of the time 2 - Moderate amount of time (3 to 4 days) 1 - Some of the time (1 to 2 days) 0 - Rarely/none of the time (<1 day) Answers of 2 or 3 to either question meet criteria for exhaustion			
Slowness	Gait speed per 15-foot distance	Slowest 20% (by sex, height)			
		Males:		Females:	
		≤7 seconds for height ≤173 cm		≤7 seconds for height ≤159 cm	
		≤6 seconds for height >173 cm		≤6 seconds for height >159 cm	
Low activity	Based on short version of the Minnesota Leisure Time Activity Questionnaire to determine Kcals per week expended	Lowest 20% (by sex) Males: <383 Kcals/week expended Females: <270 Kcals/week expended			

Frailty: ≥3 out of 5 characteristics present.

Intermediate or prefrail: 1 to 2 out of 5 characteristics present.

Robust: 0 out of 5 characteristics present.

BMI: body mass index.

Adapted from: Fried LP, Tangen CM, Walston J, et al. Frailty in older adults: Evidence for a phenotype. J Gerontol A Biol Sci Med Sci 2001; 56:M146.

Graphic 144431 Version 1.0

Figure 2: Clinical Frailty Scale

	1	Very fit	People who are robust, active, energetic and motivated. They tend to exercise regularly and are among the fittest for their age.
	2	Fit	People who have no active disease symptoms but are less fit than category 1. Often, they exercise or are very active occasionally , eg, seasonally.
	3	Managing well	People whose medical problems are well controlled , even if occasionally symptomatic, but who often are not regularly active beyond routine walking.
	4	Living with very mild frailty	Previously "vulnerable," this category marks early transition from complete independence. While not dependent on others for daily help, often symptoms limit activities . A common complaint is being "slowed up" and/or being tired during the day.
	5	Living with mild frailty	People who often have more evident slowing and need help with high-order IADLs (finances, transportation, heavy housework). Typically, mild frailty progressively impairs shopping and walking outside alone, meal preparation, and medications and begins to restrict light housework.
	6	Living with moderate frailty	People who need help with all outside activities and with keeping house . Inside, they often have problems with stairs and need help with bathing and might need minimal assistance (cuing, standby) with dressing.
	7	Living with severe frailty	Completely dependent for personal care , from whatever cause (physical or cognitive). Even so, they seem stable and not at high risk of dying (within approximately 6 months).
	8	Living with very severe frailty	Completely dependent for personal care and approaching end of life. Typically, they could not recover even from a minor illness.
	9	Terminally ill	Approaching the end of life. This category applies to people with a life expectancy <6 months , who are not otherwise living with severe frailty . (Many terminally ill people can still exercise until very close to death.)

Scoring frailty in people with dementia

The degree of frailty generally corresponds to the degree of dementia. Common **symptoms in mild dementia** include forgetting the details of a recent event, though still remembering the event itself, repeating the same question/story, and social withdrawal.

In **moderate dementia**, recent memory is very impaired, even though they seemingly can remember their past life events well. They can do personal care with prompting.

In **severe dementia**, they cannot do personal care without help.

In **very severe dementia** they are often bedfast. Many are virtually mute.

IADLs: instrumental activities of daily living.

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Figure 3: Approach to clinical management of patients with frailty

	Fit	Prefrailty	Frailty	End-stage frailty
Frailty score	- Fried frailty phenotype, 0 points - Deficit-accumulation frailty index of <0.10 - Score on Clinical Frailty Scale, 1 to 3	- Fried frailty phenotype, 1 or 2 points - Deficit-accumulation frailty index of 0.10 to <0.20 - Score on Clinical Frailty Scale, 4	- Fried frailty phenotype, 3 or 4 points - Deficit-accumulation frailty index of 0.20 to <0.55 - Score on Clinical Frailty Scale, 5 to 7	- Fried frailty phenotype, 5 points - Deficit-accumulation frailty index of ≥0.55 - Score on Clinical Frailty Scale, 8 or 9
Goal	Increase physiological reserve	Increase physiological reserve	Preserve physiological reserve and prevent avoidable stressors	Provide comfort
Lifestyle	- Exercise and physical activity - High-quality diet - Social engagement	- Exercise and physical activity - High-quality diet (protein intake) - Social engagement	- Less intense exercise may be better tolerated - High-quality diet (protein intake) - Social engagement	- Physical activity as tolerated - Diet as tolerated - Social engagement as tolerated
Disease management	Apply disease-based guidelines	Apply disease-based guidelines	Consider trade-off between disease and treatment burden	Deescalate treatments
Preventive care	- Vaccination - Cancer screening	- Vaccination - Cancer screening	- Vaccination - Individualize cancer screening (time to benefit versus remaining life expectancy)	- Vaccination - Stop cancer screening
Interventions for frailty		- Treat reversible causes of frailty - Exercise and physical activity - Nutritional counseling and supplementation - CGA and multidisciplinary intervention - Comprehensive medication review	- Treat reversible causes of frailty - Rehabilitation (PT and OT) - Nutritional counseling and supplementation - CGA and multidisciplinary intervention - Comprehensive medication review	Comprehensive medication review
Patient engagement	Patient-centered goal	Patient-centered goal	Patient-centered goal	Patient-centered goal
Social support	Social support (family and caregiver)	Social support (family and caregiver)	Social support (family and caregiver)	Social support (family and caregiver)

CGA: comprehensive geriatric assessment; OT: occupational therapy; PT: physical therapy.

From: Kim DH, Rockwood K. Frailty in older adults. *N Engl J Med* 2024; 391:538. Copyright © 2024 Massachusetts Medical Society. Reprinted with permission from Massachusetts Medical Society.

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Table 2: Sleep hygiene guidelines

Recommendation	Details
Regular bedtime and rise time	Having a consistent bedtime and rise time leads to more regular sleep schedules and avoids periods of sleep deprivation or periods of extended wakefulness during the night.
Avoid napping	Avoid napping, especially naps lasting longer than 1 hour and naps late in the day.
Limit caffeine	Avoid caffeine after lunch. The time between lunch and bedtime represents approximately 2 half-lives for caffeine, and this time window allows for most caffeine to be metabolized before bedtime.
Limit alcohol	Recommendations are typically focused on avoiding alcohol near bedtime. Alcohol is initially sedating, but activating as it is metabolized. Alcohol also negatively impacts sleep architecture.
Avoid nicotine	Nicotine is a stimulant and should be avoided near bedtime and at night.
Exercise	Daytime physical activity is encouraged, in particular, 4 to 6 hours before bedtime, as this may facilitate sleep onset. Rigorous exercise within 2 hours of bedtime is discouraged.
Keep the sleep environment quiet and dark	Noise and light exposure during the night can disrupt sleep. White noise or ear plugs are often recommended to reduce disturbing noise. Using blackout shades or an eye mask is commonly recommended to reduce light. This may also include avoiding exposure to television or technology near bedtime, as this can have an impact on circadian rhythms by shifting sleep timing later.
Bedroom clock	Avoid checking the time at night. This includes alarm clocks and other time pieces (eg, watches and smart phones). Checking the time increases cognitive arousal and prolongs wakefulness.
Evening eating	Avoid a large meal close to bedtime. Eat a healthy and filling (but not too heavy) meal in the early evening and avoid late-night snacks.

Graphic 122804 Version 5.0

Table 3: Mnemonic for common causes of malnutrition in older adults

MEALS ON WHEELS	
M	Medications
E	Emotional problems (depression)
A	Anorexia (nervosa or tardive)
L	Late-life paranoia or alcoholism
S	Swallowing disorders
O	Oral factors
N	No money
W	Wandering (in patients with dementia)
H	Hyperthyroidism, hyperparathyroidism
E	Entry problems/malabsorption
E	Eating problems (severe tremor, stroke, weakness)
L	Low-salt or low-cholesterol diets
S	Shopping and food preparation problems

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